

Instructor: Dr. Jim D. Pham
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Office Hrs: By Appointment
Web page: <http://www.ohlone.edu/jpham/>
Section: CS-124-03 (Hybrid; TTh; 6:30PM – 7:40PM)

Course Description:

This course involves the study and implementation of data structure programming techniques. The emphasis is on the data structures of stacks, queues, lists, trees, and hash tables; the use of recursion; and the application of these tools primarily to searching and sorting. Students implement these concepts by writing programs in an object-oriented language such as C++.

Prerequisites:

Advisory: CS 102 Introduction to Computer Programming Using C++

Textbook:

C++ Programming: Program Design Including Data Structures, 8th Edition
Author: D.S. Malik
ISBN-13: 978-1-337-11756-2

Student Learning Outcomes

After completing the course, students will be able to:

1. Propose and evaluate the separation of specification and implementation
2. Apply a systematic approach to the design, construction, and management of computer
3. programs, emphasizing programming style, documentation, and debugging techniques.
4. Demonstrate knowledge of the basic data structures of stacks, lists, trees, graphs, queues, and hash tables.
5. Implement (program) these structures in their appropriate applications, such as sorting and searching.
6. Compare and contrast benefits and costs of different data structures to choose the best one for a particular application.

Teaching Methods:

1. Lectures: The presentations slides, and assignments will be accessed through Ohlone Canvas. See Instructional Resources below.
2. Assignments: End of chapter activities and online activities will be assigned weekly to reinforce material in the text. These assignments may require the application of various software packages.
3. Quizzes: Weekly unannounced quizzes may be given to help ensure students keep up with assigned material.
4. Participation: Student participation will be graded by the level of class participation on the Discussion Board. Each student will be required to do the minimum of 2 posts (1 post and 1 reply) per chapter on Canvas Discussion Board.

Instructional Resources:

Canvas: Class documents, syllabus, lectures, assignments, and discussions will be posted on Canvas. You access Canvas at this URL: <https://ohlone.instructure.com>.

CANVAS CONFERENCES:

Depending on the complexity of the topics, I may conduct live lectures and labs using Canvas Conferences. Details Web Conferences will be posted on Canvas.

Email:

If you send me email, then please put **course number – name – topic** into the email **Subject** field. For example: CS 124 – Name – Issue

Discussion Board:

Your participation on the Discussion Board (i.e. Canvas) will be an important aspect of the class. Students will be required to post and respond to at least two posts on the discussion board.

Labs and Assignments:

Lab programming assignment will be assigned each week. It is the student responsibility to submit the class assignments at due date.

Exams and Tests:

There will be a) Lab Assignments; b) Online Participation; c) Quizzes; d) Midterm; e) Final Exam. You will have three attempts for each Chapter Exam and single attempts for the Midterm and Final(s). The final exam will not be comprehensive in nature. However, the instructor reserves the right to retest on material that was not appropriately comprehended. The questions will be programming, multiple-choice, true/false, and short answer fill-ins. All exams will be timed.

Policy on Student Work:

Labs and assignments are to be done individually, unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

GRADING:

Canvas Participation:	10%
Quizzes:	10%
Labs:	30%
Midterm:	20%
Final Exam:	30%

GRADING SCALE:

A = 90-100
B = 80-89
C = 65-79
D = 55-64
F = <55

Course Schedule: The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Session	Date	Activity
1	08/31/2021	Chapters 10 Classes and Data Abstraction - Chapter 10 Lab Activities
2	09/08/2021	Holiday/Labor Day (9/4-9/6) (weekend classes do not meet)
3	09/14/2021	Chapter 11 Inheritance and Composition - Chapter 11 Lab Activities
4	09/21/2021	Chapter 12 Pointers, Classes, Virtual Functions, Abstract Classes and List Chapter 12 Lab Activities
	09/28/2021	Chapter 13 Overloading and Templates - Chapter 13 Lab Activities
5	10/05/2021	Chapter 14 Exception Handling - Chapter 14 Lab Activities
6	10/12/2021	Chapter 15 Recursive - Chapter 15 Lab Activities
7	10/19/2021	Midterm (Chapter 10 – Chapter 15)
8	10/26/2021	Chapter 16 Linked Lists - Chapter 16 Lab Activities
9	11/02/2021	Chapter 17 Stacks and Queues - Chapter 17 Lab Activities
10	11/09/2021	Chapter 18 Searching and Sorting Algorithms - Chapter 18 Lab Activities
11	11/16/2021	Chapter 19 Binary Tree - Chapter 19 Lab Activities
12	11/23/2021	Chapter 20 Graph - Chapter 20 Lab Activities
14	11/29/2021	- Holiday: Thanksgiving (Ohlone College closed)
15	12/07/2021	Chapter 21 Standard Template Library (STL) - Chapter 21 Lab Activities
16	12/14/2021	Final Exam